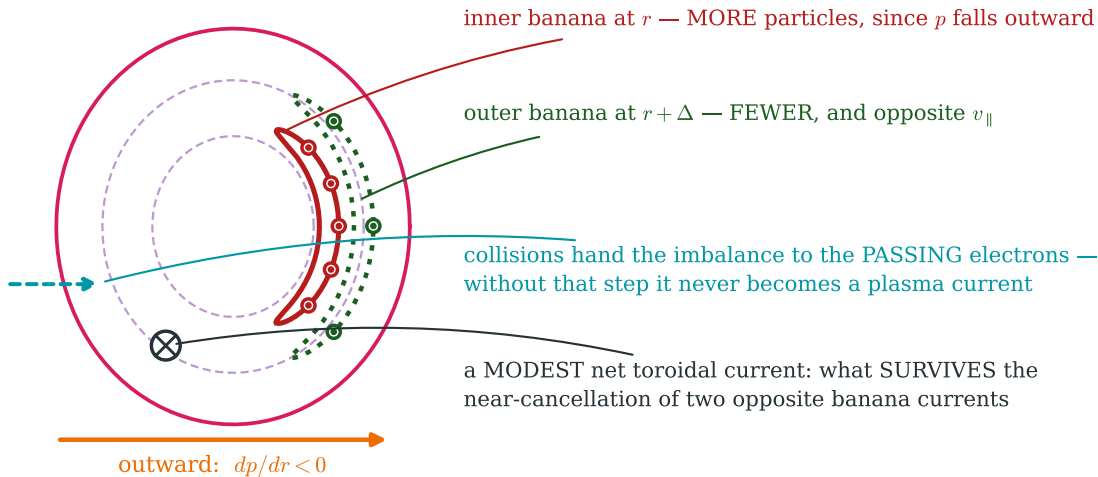


FROM TRAPPED PARTICLES TO A CURRENT — and why it is not free



1. At a given point the two adjacent bananas carry OPPOSITE $v_{\parallel}$, so their currents nearly cancel.
2. Because $dp/dr < 0$, slightly more particles sit on the inner orbit — the cancellation is INCOMPLETE.
3. Collisions transfer that residue to the passing electrons. That step is what makes it a plasma current.
4. Scaling: $j_{bs} \sim -C \frac{\sqrt{\varepsilon}}{B_0} \frac{dp}{dr}$, the $\sqrt{\varepsilon}$ inherited from the trapped fraction $\sim \sqrt{2\varepsilon/(1+\varepsilon)}$.

The coefficient C is left unwritten ON PURPOSE. It is where the kinetic model enters — collisionality regime, profile shape, impurity content — and the primer's own audit of the 2015 baseline turns on exactly that number. Quoting one here would hide the modelling choice the audit later argues about.

This is an INCOMPLETE CANCELLATION, not free current. It exists only while the pressure gradient is maintained, and maintaining that gradient is what the heating system is for.